

Apache Hadoop Installation Guide

Phase 1: Operating System Preparation

Hadoop natively runs best on Linux. Depending on your operating system, you will need to prepare a Linux environment (specifically Ubuntu) before running the installation script.

For Windows Users (WSL)

You will use the **Windows Subsystem for Linux (WSL)** to run a native Ubuntu environment directly within Windows.

1. Open **PowerShell** as an Administrator.
2. Run the following command to install WSL and Ubuntu:
- 3.

```
Shell  
wsl --install
```

4. Restart your computer if prompted.
5. Open the "Ubuntu" app from your Start Menu, and create your Linux username and password when prompted.

For Mac Users (Ubuntu VM)

Native installation on macOS (especially Apple Silicon M1/M2/M3 chips) requires complex workarounds. It is highly recommended to use an Ubuntu Virtual Machine.

1. Install a VM manager like **Multipass**, **UTM**, or **VirtualBox**.
 - *Multipass Quick Start:* Open your terminal and run `brew install multipass`, then launch an instance with `multipass launch --name hadoop-vm --cpus 2 --disk 20G --memory 4G`.
 - *Access the VM:* Run `multipass shell hadoop-vm`.
2. Once inside your Ubuntu terminal, proceed to Phase 2.

Phase 2: The Automated Installation Script

Instead of manually editing configuration files, we have prepared a single Bash script that will download Hadoop, configure Java, set up your environment variables, and prepare the necessary XML files for your current user.

1. Create the installation script file:

In your Ubuntu/WSL terminal, run the following command to open a text editor:

```
Shell
nano install_hadoop.sh
```

2. Copy and Paste the Script:

Copy the code block below and paste it into the nano editor (Right-click to paste in WSL).

```
Shell
#!/bin/bash

# Define Versions and Paths
HADOOP_VERSION="3.3.6"
HADOOP_TAR="hadoop-$HADOOP_VERSION.tar.gz"
HADOOP_URL="https://downloads.apache.org/hadoop/common/hadoop-$HADOOP_VERSION/$HADOOP_TAR"
JAVA_PATH="/usr/lib/jvm/java-8-openjdk-amd64"

echo "--- Starting Automated Hadoop $HADOOP_VERSION Installation ---"

# 1. Detect Shell Profile (Dynamic detection)
CURRENT_SHELL=$(basename "$SHELL")
if [ "$CURRENT_SHELL" = "zsh" ]; then
    SHELL_PROFILE="$HOME/.zshrc"
elif [ "$CURRENT_SHELL" = "bash" ]; then
    SHELL_PROFILE="$HOME/.bashrc"
else
    SHELL_PROFILE="$HOME/.profile"
```

```

fi
echo "Detected shell: $CURRENT_SHELL. Environment variables will
be saved to $SHELL_PROFILE"

# 2. Install Dependencies (Java and SSH)
echo "Installing Java 8 and SSH..."
sudo apt update
sudo apt install -y openjdk-8-jdk wget openssh-server
openssh-client

# 3. Setup Passwordless SSH for Localhost
echo "Configuring SSH..."
if [ ! -f ~/.ssh/id_rsa ]; then
    ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
    cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
    chmod 0600 ~/.ssh/authorized_keys
fi

# 4. Download and Extract Hadoop
echo "Downloading Hadoop..."
if [ ! -f "$HADOOP_TAR" ]; then
    wget $HADOOP_URL
fi

echo "Extracting files (this may take a minute)..."
sudo tar -xzf $HADOOP_TAR -C /usr/local/
sudo mv /usr/local/hadoop-$HADOOP_VERSION /usr/local/hadoop
sudo chown -R $USER:$USER /usr/local/hadoop

# 5. Configure Environment Variables (Fixed Variable Expansion
Bug)
echo "Configuring Environment Variables..."
cat <<EOT >> "$SHELL_PROFILE"

# Hadoop Env Vars
export JAVA_HOME=$JAVA_PATH
export HADOOP_HOME=/usr/local/hadoop
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
EOT

# 6. Configure Hadoop XML Files (Updated to modern 3.x property
names)
echo "Writing Configuration Files..."

```

```
CONF_DIR="/usr/local/hadoop/etc/hadoop"
```

```
# hadoop-env.sh
```

```
echo "export JAVA_HOME=$JAVA_PATH" >> $CONF_DIR/hadoop-env.sh
```

```
# core-site.xml
```

```
cat <<EOT > $CONF_DIR/core-site.xml
```

```
<configuration>
```

```
<property><name>hadoop.tmp.dir</name><value>/home/$USER/tmpdata</value></property>
```

```
<property><name>fs.defaultFS</name><value>hdfs://localhost:9000</value></property>
```

```
</configuration>
```

```
EOT
```

```
# hdfs-site.xml
```

```
cat <<EOT > $CONF_DIR/hdfs-site.xml
```

```
<configuration>
```

```
<property><name>dfs.namenode.name.dir</name><value>/home/$USER/dfsdata/namenode</value></property>
```

```
<property><name>dfs.datanode.data.dir</name><value>/home/$USER/dfsdata/datanode</value></property>
```

```
<property><name>dfs.replication</name><value>1</value></property></configuration>
```

```
EOT
```

```
# mapred-site.xml
```

```
cat <<EOT > $CONF_DIR/mapred-site.xml
```

```
<configuration>
```

```
<property><name>mapreduce.framework.name</name><value>yarn</value>></property>
```

```
</configuration>
```

```
EOT
```

```
# yarn-site.xml
```

```
cat <<EOT > $CONF_DIR/yarn-site.xml
```

```
<configuration>
```

```
<property><name>yarn.nodemanager.aux-services</name><value>mapred
uce_shuffle</value></property>

<property><name>yarn.nodemanager.aux-services.mapreduce.shuffle.c
lass</name><value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>

<property><name>yarn.resourcemanager.hostname</name><value>localh
ost</value></property>
</configuration>
EOT
```

7. Create Local Data Directories

```
echo "Creating Data Directories..."
mkdir -p /home/$USER/tmpdata
mkdir -p /home/$USER/dfsdata/namenode
mkdir -p /home/$USER/dfsdata/datanode
```

8. Source the profile for immediate use

```
echo "Sourcing the shell profile..."
source "$SHELL_PROFILE"

echo "--- Installation Complete! ---"
echo "IMPORTANT: To ensure all paths are active in your current
window, run:"
echo "source $SHELL_PROFILE"
echo ""
echo "Then, format your namenode before starting Hadoop:"
```

3. Save and Exit: Press CTRL+O to save, ENTER to confirm, and CTRL+X to exit nano.

4. Make the script executable and run it:

```
Shell
chmod +x install_hadoop.sh
./install_hadoop.sh
```

(Note: It may ask for your password to install Java and system updates. Type it in and press Enter).

Phase 3: Initializing and Starting the Cluster

Once the script finishes successfully, you need to perform three final commands to start your cluster.

1. Apply the environment variables to your current session:

Shell

```
source ~/.bashrc
```

2. Format the NameNode:

This initializes the HDFS file system. **(Only do this once!)**

Shell

```
hdfs namenode -format
```

3. Start all Hadoop Services (HDFS and YARN):

Shell

```
start-dfs.sh  
start-yarn.sh
```

4. Verify the Services:

Run the Java Process Status tool to ensure everything is running:

```
Shell  
jps
```

Success! If you see NameNode, DataNode, SecondaryNameNode, ResourceManager, and NodeManager in the output list, your Hadoop installation is fully operational!

Tip: You can view your cluster visually by opening a web browser on your host machine and going to <http://localhost:9870>.